

KONYS, INC.

LCD Module Procurement Strategy

Monte Carlo Simulation Analysis - Assignment 2

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Konys, Inc. Executive Board

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Executive Summary

This report analyzes procurement strategies for LCD modules used in MC20 mobile phone production. We used Monte Carlo simulation with 10,000 iterations to evaluate long-term purchase contracts and option contracts under demand uncertainty and volatile spot prices.

Key Findings

Best Strategy (Recommended): Purchase 1,200,000 units + Option 1,900,000 units

- Expected quarterly profit: **\$9,251,007**
- 90% confidence interval: [\$8,860,444, \$9,641,571]
- Improvement: **30.9%** over purchase-only strategy

Recommendation

We recommend adopting the combined purchase and option contract strategy with:

- Purchase contract: 1,200,000 units at \$17.00/unit = \$20,400,000
- Option contract: 1,900,000 units at \$0.50 reservation + \$16.75 exercise = \$950,000
- Total upfront cost: \$21,350,000

This strategy provides flexibility to respond to demand uncertainty while protecting against spot price volatility.

1 Introduction

1.1 Problem Context

Konys, Inc. faces challenges in procuring LCD modules for the MC20 mobile phone due to uncertain demand and volatile spot prices. Historical Q3 demand ranges from 872,841 to 3,246,210 units, while spot prices fluctuate between \$12.54 and \$26.97.

1.2 Procurement Options

Two procurement strategies are available:

1. **Long-term Purchase Contract:** Commit to fixed quantities at \$17.00/unit
2. **Option Contract:** Reserve the right to purchase at \$16.75/unit for \$0.50/unit upfront

1.3 Methodology

We used Monte Carlo simulation to evaluate different strategies:

- Fitted probability distributions to historical data (Normal for demand, Beta for spot prices)
- Ran 10,000 simulation iterations per scenario
- Tested order quantities from 1.0M to 3.6M units in 100K increments
- Calculated expected profits, confidence intervals, and risk metrics

2 Part A: Purchase Contract Analysis

2.1 A1 & A2: Model Development and Distribution Fitting

We developed a profit calculation model that accounts for revenue, salvage value, LCD costs, production costs, and fixed expenses (\$55,757,500 per quarter).

From historical data, we fitted:

- **Q3 Demand:** Normal distribution with mean = 2,159,071 and std dev = 751,724 units
- **Spot Prices:** Beta distribution with parameters $\alpha = 0.9417$, $\beta = 0.8548$

2.2 A3: Expected Profit at 2,170,000 Units

With an order quantity of 2,170,000 units:

Table 1: Results for A3 (Order = 2,170,000 units)

Metric	Value
Expected Profit	\$5,869,699
Standard Deviation	\$26,359,926
10th Percentile	-\$30,403,473
50th Percentile	\$10,234,493
90th Percentile	\$36,572,690

The order quantity of 2.17M is close to average demand (2.16M) but shows high variability and significant downside risk.

2.3 A4: Optimal Order Quantity

After testing order quantities from 1.0M to 3.6M units, we found:

Table 2: Optimal Purchase Contract (A4)

Parameter	Value
Optimal Order Quantity	1,200,000 units
Expected Profit	\$7,065,764
Standard Deviation	\$22,153,737
90% Confidence Interval	[\$6,701,315, \$7,430,212]
Improvement over A3	+20.4%

The optimal order (1.2M) is conservative, which limits excess inventory risk while relying on spot market for demand surges.

3 Part B: Option Contract Analysis

3.1 B1: Option Contract Model

The option contract allows Konys to reserve capacity at \$0.50/unit (paid upfront) and exercise the option at \$16.75/unit only when spot prices exceed this level. This provides flexibility without full commitment.

3.2 B2: Optimal Option Quantity (Order = 2,170,000)

With order quantity fixed at 2,170,000 units:

Table 3: Optimal Option Strategy with Order = 2.17M (B2)

Parameter	Value
Order Quantity	2,170,000 units
Optimal Option Quantity	1,200,000 units
Expected Profit	\$5,801,416
90% Confidence Interval	[\$5,344,885, \$6,257,948]

This strategy performs slightly worse than A3 due to the \$600K reservation fee and non-optimized order quantity.

3.3 B3: Risk Comparison (A3 vs B2)

Both strategies use 2,170,000 units order quantity:

Table 4: Comparison at Order = 2.17M units

Metric	A3 (Purchase)	B2 (Option)
Expected Profit	\$5,869,699	\$5,801,416
Standard Deviation	\$26,359,926	\$27,751,180
10th Percentile	-\$30,403,473	-\$31,866,361
50th Percentile	\$10,234,493	\$8,809,011
90th Percentile	\$36,572,690	\$39,651,104

Analysis:

- B2 has slightly lower expected profit (-1.2%) due to reservation fee
- B2 performs worse at lower percentiles but better at higher percentiles
- At 90th percentile, B2 is \$3.08M higher (+8.4%)
- Option provides protection against high spot prices in demand surge scenarios

3.4 B4: Optimal Combined Strategy (Our recommendation)

Using the optimal order from A4 (1,200,000 units) with optimized option quantity:

Table 5: Optimal Combined Strategy (B4)

Parameter	Value
Order Quantity	1,200,000 units
Optimal Option Quantity	1,900,000 units
Total Capacity	3,100,000 units
Expected Profit	\$9,251,007
Standard Deviation	\$23,741,179
90% Confidence Interval	[\$8,860,444, \$9,641,571]

Comparison with A4:

Table 6: B4 vs A4 Performance

Metric	A4 (Purchase)	B4 (Option)
Expected Profit	\$7,065,764	\$9,251,007
Improvement	-	+\$2,185,243 (+30.9%)
10th Percentile	-\$18,450,066	-\$19,887,686
50th Percentile	\$6,716,040	\$9,684,615
90th Percentile	\$34,809,739	\$39,488,449

Why B4 Yields More Profit:

The combined strategy achieves 30.9% higher profit through:

1. **Conservative base order (1.2M):** Limits excess inventory risk while staying below average demand
2. **Upside protection:** When demand is high and spot prices spike, the option allows purchase at \$16.75 instead of spot prices up to \$27
3. **Example:** If demand = 3.5M units and spot = \$25:
 - A4 buys 2.3M at \$25 = \$57.5M additional LCD cost
 - B4 exercises 1.9M option at \$16.75 + buys 0.4M at \$25 = \$41.825M
 - Savings: \$15.675M in this scenario
4. **Bounded downside:** Maximum additional cost is \$950K reservation fee (known and limited)

5. **Asymmetric payoff:** Small fixed cost (\$950K) vs. large potential savings (up to \$19M when spot spikes)

The option creates favorable asymmetry where the maximum downside (\$950K) is small compared to the potential upside savings.

4 Analysis and Recommendations

4.1 Strategy Comparison

Table 7: Summary of All Strategies

Strategy	Expected Profit	Rank
B4: Optimal Combined (1.2M + 1.9M opt)	\$9,251,007	1st
A4: Optimal Purchase (1.2M)	\$7,065,764	2nd
A3: Given Purchase (2.17M)	\$5,869,699	3rd
B2: Given Combined (2.17M + 1.2M opt)	\$5,801,416	4th

4.2 Financial Impact

Strategy B4 delivers a quarterly profit increase of \$2,185,243 over the best purchase-only alternative (A4), representing a 30.9% improvement. The \$950,000 option reservation fee generates \$2.19 million in additional profit—a 130% ROI in a single quarter. If applied across all four quarters of MC20 production, the annual profit improvement would be approximately \$8.74 million.

4.3 Risk Profile

B4 provides better risk-adjusted returns than A4, with lower risk per unit of return (coefficient of variation: 2.57 vs 3.13). The strategy outperforms A4 in approximately 75% of scenarios, particularly when demand is medium to high. It only underperforms in very low demand cases (10th percentile) where the \$950,000 reservation fee becomes a burden. However, this represents a known, bounded cost compared to unlimited spot price exposure.

4.4 Implementation Plan

Implementation follows three phases. First, we negotiate two contracts with Paravan Systems: a purchase contract for 1,200,000 units at \$17.00/unit and an option contract for 1,900,000 units (\$0.50 reservation, \$16.75 exercise), requiring \$21,350,000 upfront commitment.

Second, in mid-June 2013, we monitor demand indicators and spot prices to decide whether to exercise the option. The decision rule is straightforward: exercise if forecasted demand exceeds 1.2 million units and spot price exceeds \$16.75.

Third, in July 2013 we receive the 1.2 million units from the purchase contract plus any exercised option units. Spot market is only needed if demand exceeds 3.1 million units (less than 5% probability).

4.5 Supplier Benefits

This strategy benefits Paravan Systems with guaranteed revenue of \$21.35 million and potential maximum of \$53.175 million if the full option is exercised. The advance commitment provides better production planning visibility and addresses their previous concerns about order volatility from the MC19 cycle, strengthening the partnership for future collaboration.

5 Conclusions

Our analysis demonstrates that a combined purchase and option contract strategy significantly outperforms pure purchase approaches. The recommended strategy (B4) with 1,200,000 units purchase and 1,900,000 units option delivers:

- **Expected quarterly profit: \$9,251,007**
- **30.9% improvement over best purchase-only strategy**
- **Superior risk-adjusted returns**
- **Strong ROI: 130% on option investment**

The strategy balances committed supply with operational flexibility, allowing Konys to respond effectively to demand uncertainty while protecting against spot price volatility. The \$950,000 option investment is justified by the \$2.19M expected incremental profit.

We recommend immediate adoption of this strategy for Q3 2013 and future quarters.

Appendices

Appendix A: Model Validation

We validated our model against provided test scenarios:

Table 8: Model Validation Results

Test	Demand	Order	Option	Spot	Expected	Status
1	3.0M	2.17M	1.0M	\$24	\$35,450,000	PASS
2	4.0M	2.17M	1.0M	\$24	\$60,182,500	PASS
3	4.0M	2.17M	1.0M	\$12	\$74,892,500	PASS

All validation tests passed with 100% accuracy.

Appendix B: Technical Details

Distribution Parameters:

- Demand: Normal($\mu = 2,159,071$, $\sigma = 751,724$)
- Spot Price: Beta($\alpha = 0.9417$, $\beta = 0.8548$, location = \$12.54, scale = \$14.43)

Simulation Settings:

- Number of iterations: 10,000 per scenario
- Random seed: 1 (for reproducibility)
- Optimization search: 1.0M to 3.6M units in 100K increments

Key Assumptions:

- Historical patterns representative of future
- No supply disruptions or quality issues
- Fixed expenses constant across strategies (\$55,757,500/quarter)
- Option exercise decision based on observed demand signals

Appendix C: Cost Parameters

Table 9: Contract and Cost Parameters

Parameter	Value
Purchase Contract Price	\$17.00/unit
Option Reservation Fee	\$0.50/unit
Option Exercise Price	\$16.75/unit
Salvage Value	\$5.30/unit
Selling Price	\$150.00/unit
Other Production Costs	\$102.50/unit
Fixed Quarterly Expenses	\$55,757,500
SG&A	\$37,920,000
Design & Research	\$17,837,500